

Here are five systematic troubleshooting steps I would take to diagnose why a laptop won't power on, even when plugged in:

1. Check the power source and AC adapter

- What I'm checking for: Whether the laptop is actually receiving power.
- Verify that the wall outlet works by plugging in another device.
- Inspect the charger for frayed cables, bent connectors, or damage.
- If possible, test with a known-good compatible charger to rule out a faulty power adapter.

2. Inspect the charging port and power indicators

- What I'm checking for: Signs that power is reaching the laptop.
- Look for charging LEDs or status lights when the adapter is connected.
- Check if the charging port is loose, damaged, or obstructed by debris, which could prevent a proper electrical connection.

3. Perform a power reset (hard reset)

- What I'm checking for: Whether the laptop is stuck due to residual electrical charge or a firmware issue.
- Disconnect the charger, remove the battery (if removable), hold the power button for 15–30 seconds, then reconnect the charger and try powering it on again.
- If the laptop starts afterward, the issue was likely related to a temporary power state rather than failed hardware.

4. Test the battery

- What I'm checking for: Whether a failed battery is preventing startup.
- If the battery is removable, remove it and try powering on using only the AC adapter.

- If the laptop powers on without the battery, the battery may be defective. If it still does not power on, the problem likely lies elsewhere.

5. Check for hardware or motherboard issues

- What I'm checking for: Evidence of internal hardware failure.
- Listen for fan noise, beeps, or observe keyboard backlights when pressing the power button.
- Disconnect external devices (USB drives, docking stations, etc.) that could interfere with startup.
- If there are no signs of power despite a working charger and outlet, the issue may involve the motherboard, DC power jack, RAM, or another internal component requiring further hardware diagnostics or repair.